

Advanced Natural Language Processing

Discussion 3: lecture 4: Quantitative Measurements and Tutorial 03

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Learning Goals Today

- Understand how to measure the quality of NLP algorithms.
- Experience hands-on the theory of this lecture



Why Quantitative Measurements of NLP Algorithms?

Because **we need objective evidence** that an NLP system works as intended.

- **Avoid subjectivity**

Human impressions (“it seems good”) are unreliable.

Quantitative metrics (accuracy, F1, BLEU, perplexity, etc.) give *measurable* performance.

- **Compare algorithms fairly**

Without numbers, we can’t say whether model A is better than model B.

Quantitative evaluation allows benchmarking on shared datasets.

- **Track improvements over time**

NLP development is iterative.

Metrics let us see if a change improves or harms performance.

- **Generalize beyond examples**

A model may look great on a few sample sentences but fail on thousands of unseen cases.

Metrics computed over large datasets measure robustness and generalizability.

- **Make decisions with stakeholders**

In real-world products, decisions need justification:

“We should deploy this model because it reduces error by 12%.”



Ethical Considerations to use Quantitative Measurements

- **Detect Bias & Ensure Fairness**
Measure performance across groups to prevent discrimination.
- **Guarantee Safety & Reliability**
Quantify error rates to avoid harmful failures in sensitive contexts.
- **Enable Accountability & Transparency**
Numbers make model decisions auditable and explainable.
- **Prevent Overconfidence & Misuse**
Evaluate uncertainty so humans know when to intervene.

Exercise: Precision and Recall

Question: An IR system returns 25 relevant documents, and 15 non-relevant documents. There are a total of 50 relevant documents in the collection.

- a. What is the precision of the system on this search.
- b. What is its recall.
- c. What is the f_1 value.
- d. How can we create a f value that prefers recall over precision.
- e. The Kappa value of the data set against which these queries are tested is lower than 60%, what does this say about the quality of the data set.

Exercise: Precision and Recall

- Why is accuracy not a good quantitative measurement for information retrieval problems.
- Explain why an 11-points recall-precision graph is the best method to measure the quality of an information retrieval system.
- Explain how it can be derived.

High accuracy does not mean you find something useful

- You can build a 99 % accurate search engine on a low budget....

snoogle.com

Search for:

0 matching results found.

- Caused because in IR there is often a **large imbalance** between relevant documents (**very small**) and non-relevant document (**very large**).
- This is why we need to measure both **precision** and **recall**.

Is Perplexity still a useful measurement?

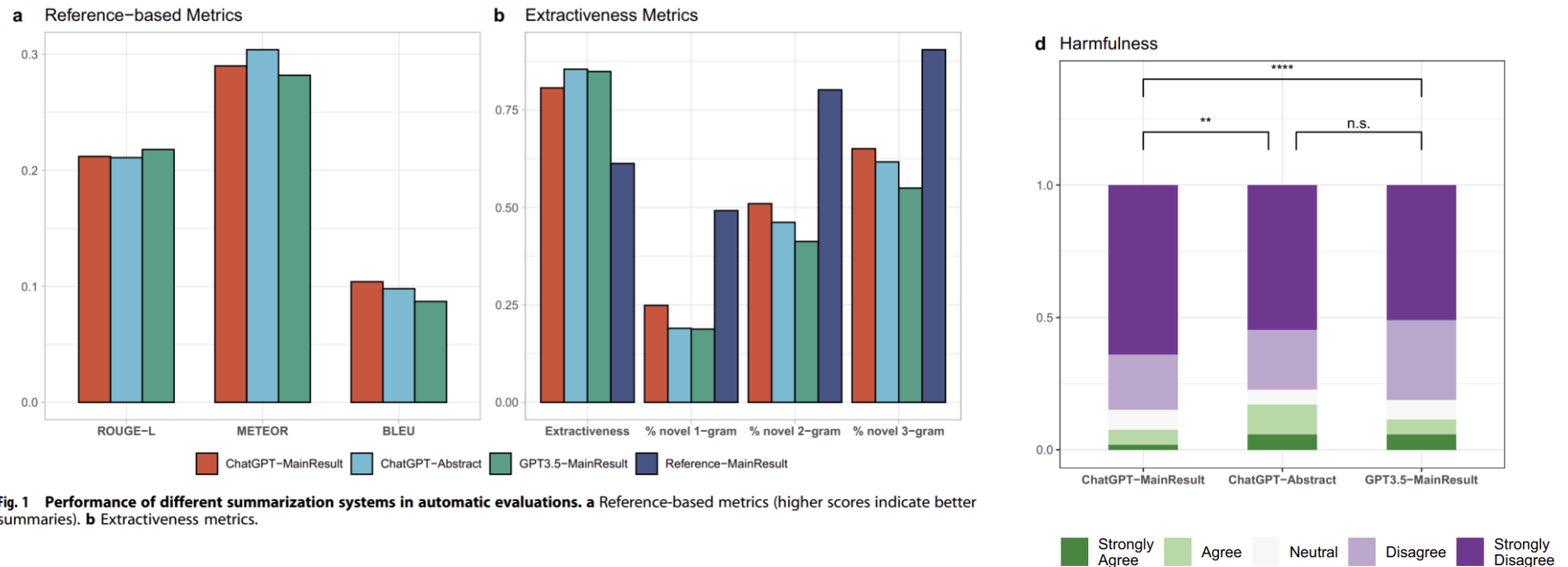
- GPT-2 and LLaMA-2 publish perplexity because they are *pretrained language models*.
- GPT-5, Gemini 2, Claude 3, etc., do **not** publish perplexity because they are *instruction-tuned chat models*.
- If measured, GPT-5 / Gemini would likely score **below 3** — significantly better than LLaMA-2 and GPT-2.



Modern frontier models (GPT-4, GPT-5, Gemini 1.5/2.0, Claude 3, etc.) do not publish perplexity for three reasons:

1. RLHF / instruction tuning breaks perplexity comparability. Perplexity is defined for next-token prediction on a fixed language-model objective. Once a model is instruction-tuned (SFT + RLHF), it is no longer a pure next-token model, so perplexity loses meaning across models.
2. Companies benchmark on tasks instead (MMLU, GSM8K, HumanEval, etc.)
 - GPT-5 reports improvements on accuracy and hallucination rates, not perplexity.
 - Google Gemini and Anthropic Claude do the same.

Rouge can be high, but depending on what is left out or added, the quality can really differ!



Data Structure Requirements

- **Predictions:** A list of dictionaries containing the question `id`, the `prediction_text`, and a `no_answer_probability` (often used to tune thresholds for unanswerable questions).
- **References:** A list of dictionaries containing the question `id` and an `answers` dictionary. The `answers` field must include a list of possible correct `text` strings and their corresponding `answer_start` character indices from the context.

For large datasets, you can also run the official standalone script, `evaluate-v2.0.py`, by passing the path to the ground truth JSON file and your model's prediction file.

```
# Define sample ground truth (references)
# Unanswerable questions have empty 'text' and 'answer_start' lists
references = [
    {
        "id": "q1_answerable",
        "answers": {
            "text": ["1976", "the year 1976"], # Multiple valid answers allowed
            "answer_start": [97, 92]
        }
    },
    {
        "id": "q2_unanswerable",
        "answers": {
            "text": [],
            "answer_start": []
        }
    }
]

# Define sample model outputs (predictions)
predictions = [
    {
        "id": "q1_answerable",
        "prediction_text": "1976",
        "no_answer_probability": 0.0
    },
    {
        "id": "q2_unanswerable",
        "prediction_text": "", # Empty string for unanswerable questions
        "no_answer_probability": 0.9 # High probability of no answer
    }
]

# Compute scores
results = squad_v2_metric.compute(predictions=predictions, references=references)

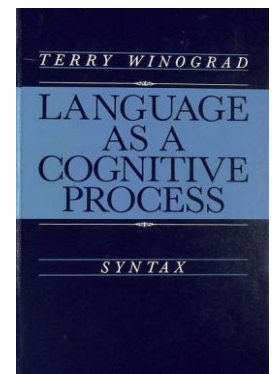
# Print specific metrics
print(f"Exact Match: {results['exact']}%")
print(f"F1 Score: {results['f1']}")
print(f"HasAns F1: {results['HasAns_f1']}")
print(f"NoAns F1: {results['NoAns_f1']}")
```

Modern Approaches for Semantic Matching

- **Embedding-Based Similarity:** Tools like BERTScore and SimCSE use deep learning to transform text into mathematical vectors (embeddings). This allows the system to recognize that "The sun is hot" and "The star's temperature is extremely high" are semantically similar despite having zero word overlap.
- **Claim-Level Decomposition:** Rather than comparing two long sentences as a whole, break an answer into distinct factual claims.
- **The RAG-Trias:** Faithfulness, Answer Relevance, Context Relevance
- **Atomic Fact checking** (with text-mining analysis)
- **Natural Language Inference (NLI):** This method uses a separate model to determine if the source document "entails" (supports), "contradicts," or is "neutral" toward the generated answer.
- **LLM as a judge:** An LLM judge can follow a detailed rubric to score answers on dimensions like comprehensiveness, conciseness, and clarity, which token-level metrics cannot assess.

Winograd schema challenge

- The Winograd schema challenge (WSC) is a test of machine intelligence proposed by Hector Levesque, a computer scientist at the University of Toronto. Designed to be an improvement on the Turing test, it is a multiple-choice test that employs questions of a very specific structure: they are instances of what are called Winograd schemas, named after Terry Winograd, professor of computer science at Stanford University.
- On the surface, Winograd schema questions simply require the resolution of anaphora: the machine must identify the antecedent of an ambiguous pronoun in a statement. This makes it a task of natural language processing, but Levesque argues that for Winograd schemas, the task requires the use of knowledge and commonsense reasoning.



Winograd Scheme Example

“Dave told everyone in school that he wants to be a guitarist, because he thinks it is a great sounding instrument.”

Short of accessing knowledge structures that somehow encode the commonsense facts:

- that a-guitarist-plays-a-guitar, and
- that a-guitar-is-a-musical-instrument,

How to resolve “it” in since the correct referent is not even in the data, but is an object that is implicitly assumed by the semantic and cognitive content of the text.

World History

The invaders had brought in wheat and other Eurasian and African grains; peach, pear, orange, and lemon trees; chick-peas, grape vines, melons, onions, radishes, and much more. A Spanish nobleman come to America could require his Indians to furnish his table with the fruits of his ancestors.

What was the impact of this process on Europe?

- A. The population increased due to the increased diversity of crops.
- B. The population declined as a result of imported disease.
- C. The population increased due to New World immigrants.
- D. The population declined because of famine caused by the mass export of crops.

Answer: A

Task	Tested Concepts	Supercategory
Abstract Algebra	Groups, rings, fields, vector spaces, ...	STEM
Anatomy	Central nervous system, circulatory system, ...	STEM
Astronomy	Solar system, galaxies, asteroids, ...	STEM
Business Ethics	Corporate responsibility, stakeholders, regulation, ...	Other
Clinical Knowledge	Spot diagnosis, joints, abdominal examination, ...	Other
College Biology	Cellular structure, molecular biology, ecology, ...	STEM
College Chemistry	Analytical, organic, inorganic, physical, ...	STEM
College Computer Science	Algorithms, systems, graphs, recursion, ...	STEM
College Mathematics	Differential equations, real analysis, combinatorics, ...	STEM
College Medicine	Introductory biochemistry, sociology, reasoning, ...	Other
College Physics	Electromagnetism, thermodynamics, special relativity, ...	STEM
Computer Security	Cryptography, malware, side channels, fuzzing, ...	STEM
Conceptual Physics	Newton's laws, rotational motion, gravity, sound, ...	STEM
Econometrics	Volatility, long-run relationships, forecasting, ...	Social Sciences
Electrical Engineering	Circuits, power systems, electrical drives, ...	STEM
Elementary Mathematics	Word problems, multiplication, remainders, rounding, ...	STEM
Formal Logic	Propositions, predicate logic, first-order logic, ...	Humanities
Global Facts	Extreme poverty, literacy rates, life expectancy, ...	Other
High School Biology	Natural selection, heredity, cell cycle, Krebs cycle, ...	STEM
High School Chemistry	Chemical reactions, ions, acids and bases, ...	STEM
High School Computer Science	Arrays, conditionals, iteration, inheritance, ...	STEM
High School European History	Renaissance, reformation, industrialization, ...	Humanities
High School Geography	Population migration, rural land-use, urban processes, ...	Social Sciences
High School Gov't and Politics	Branches of government, civil liberties, political ideologies, ...	Social Sciences
High School Macroeconomics	Economic indicators, national income, international trade, ...	Social Sciences
High School Mathematics	Pre-algebra, algebra, trigonometry, calculus, ...	STEM
High School Microeconomics	Supply and demand, imperfect competition, market failure, ...	Social Sciences
High School Physics	Kinematics, energy, torque, fluid pressure, ...	STEM
High School Psychology	Behavior, personality, emotions, learning, ...	Social Sciences
High School Statistics	Random variables, sampling distributions, chi-square tests, ...	STEM
High School US History	Civil War, the Great Depression, The Great Society, ...	Humanities
High School World History	Ottoman empire, economic imperialism, World War I, ...	Humanities
Human Aging	Senescence, dementia, longevity, personality changes, ...	Other
Human Sexuality	Pregnancy, sexual differentiation, sexual orientation, ...	Social Sciences
International Law	Human rights, sovereignty, law of the sea, use of force, ...	Humanities
Jurisprudence	Natural law, classical legal positivism, legal realism, ...	Humanities
Logical Fallacies	No true Scotsman, base rate fallacy, composition fallacy, ...	Humanities
Machine Learning	SVMs, VC dimension, deep learning architectures, ...	STEM
Management	Organizing, communication, organizational structure, ...	Other
Marketing	Segmentation, pricing, market research, ...	Other
Medical Genetics	Genes and cancer, common chromosome disorders, ...	Other
Miscellaneous	Agriculture, Fermi estimation, pop culture, ...	Other
Moral Disputes	Freedom of speech, addiction, the death penalty, ...	Humanities
Moral Scenarios	Detecting physical violence, stealing, externalities, ...	Humanities
Nutrition	Metabolism, water-soluble vitamins, diabetes, ...	Other
Philosophy	Skepticism, phronesis, skepticism, Singer's Drowning Child, ...	Humanities
Prehistory	Neanderthals, Mesoamerica, extinction, stone tools, ...	Humanities
Professional Accounting	Auditing, reporting, regulation, valuation, ...	Other
Professional Law	Torts, criminal law, contracts, property, evidence, ...	Humanities
Professional Medicine	Diagnosis, pharmacotherapy, disease prevention, ...	Other
Professional Psychology	Diagnosis, biology and behavior, lifespan development, ...	Social Sciences
Public Relations	Media theory, crisis management, intelligence gathering, ...	Social Sciences
Security Studies	Environmental security, terrorism, weapons of mass destruction, ...	Social Sciences
Sociology	Socialization, cities and community, inequality and wealth, ...	Social Sciences
US Foreign Policy	Soft power, Cold War foreign policy, isolationism, ...	Social Sciences
Virology	Epidemiology, coronaviruses, retroviruses, herpesviruses, ...	Other
World Religions	Judaism, Christianity, Islam, Buddhism, Jainism, ...	Humanities

Table 2: Summary of all 57 tasks.

Beyond MMLU ...

While MMLU (Massive Multitask Language Understanding) is a very challenging and diverse benchmark covering 57 academic and professional tasks, there are a few datasets and benchmarks that are considered even more challenging in specific ways, particularly in testing reasoning, robustness, truthfulness, or multi-step problem solving.

Dataset	Focus	What makes it more challenging than MMLU
BBH	Broad reasoning tasks	Curated for tasks LLMs struggle with
ARC-Challenge	Science reasoning	Harder questions from standardized exams
TruthfulQA	Truthfulness, adversarial QA	Designed to cause hallucinations
GSM8K / MATH	Multi-step math reasoning	Requires structured, chain-of-thought
HellaSwag	Commonsense reasoning	Includes adversarial distractors

The Ragas Framework combines all of the above:

Ragas is a framework that helps you evaluate your Retrieval Augmented Generation (RAG) pipelines. RAG denotes a class of LLM applications that use external data to augment the LLM's context. There are existing tools and frameworks that help you build these pipelines but evaluating it and quantifying your pipeline performance can be hard. This is where Ragas (RAG Assessment) comes in.

Hint

Ground truth: Einstein was born in 1879 in Germany.

High answer correctness: In 1879, Einstein was born in Germany.

Low answer correctness: Einstein was born in Spain in 1879.

Also includes code
to generate part of
the data set.

<https://docs.ragas.io/>

```
from datasets import Dataset
from ragas.metrics import faithfulness, answer_correctness
from ragas import evaluate

data_samples = {
    'question': ['When was the first super bowl?', 'Who won the most super bowl?'],
    'answer': ['The first superbowl was held on Jan 15, 1967', 'The most super bowl won by the Pittsburgh Steelers'],
    'ground_truth': ['The first superbowl was held on January 15, 1967', 'The Pittsburgh Steelers won the most super bowl']
}

dataset = Dataset.from_dict(data_samples)
score = evaluate(dataset, metrics=[answer_correctness])
score.to_pandas()
```



LLM as a Judge: Agentic model based evaluation

One of the most popular advanced methods is using a highly capable Large Language Model (like GPT-4 or Gemini) to act as an automated evaluator.

- **Criteria-Based Rubrics:** An LLM judge can follow a detailed rubric to score answers on dimensions like comprehensiveness, conciseness, and clarity, which token-level metrics cannot assess.
- **Pairwise Comparison:** Two different model responses are presented to a "judge" LLM, which determines which one is better based on specific quality guidelines.
- **RAGAS and DeepEval:** Frameworks like RAGAS automate these multi-step factual checks by prompting an LLM to extract claims from an answer and compare them directly to the source material to calculate a "Faithfulness" score.

We will take a closer look how to combine Re-Ranking, RAG, and Agentic models later.



Example: How does this work in the Conversational Search, RAG & Agentic Tutorial

In the RAG tutorial, we move beyond just "finding information" to ensuring the generated response is grounded and truthful. We apply quantitative measures to detect hallucinations using the following methods:

1. Measuring "Faithfulness" with DSPy & LLM-as-a-Judge

- a. What you will do: Use the Semantic F1 metric and custom LLM-as-a-Judge prompts within the **DSPy** framework.
- b. How it works: We compare the generated `response` against the retrieved `context`.
- c. The Goal: Ensure that every claim in the answer is explicitly supported by the retrieved chunks. If the model introduces external (and potentially false) information not found in the context, it is flagged as a Hallucination.

2. Answer Relevance vs. Contextual Grounding

- a. Metric: We measure Answer Correctness by comparing the model's output to a "Gold Standard" human response.
- b. Detection: By analyzing the Context Relevancy, we determine if the retriever failed (noise) or if the generator "made things up" despite having the right context.

3. RAG Evaluation Cycle

- a. Retrieve: Fetch snippets using BM25 or ColBERT.
- b. Generate: Use a `ChainOfThought` module to reason through the facts.
- c. Validate: Apply the Semantic F1 metric to verify that the reasoning and the final answer are factually consistent with the source documents.
- d. Use Agen



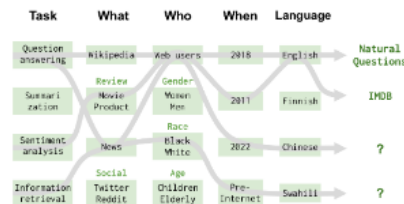
Evaluation Methods for different NLP Tasks

Task	Type	Methods
Classification (NER, Sentiment, POS, ...)	Balanced	Accuracy
	Un-balanced	Precision, Recall, F-1, 11 Points PR, ROC
Abstracting		ROUGE, METEOR, BLEU
Machine Translation		BLEU, METEOR
Dialogue / Q&A		SQuAD, MMLU
Quality of a Language Model		Perplexity
Language “Understanding” and Linguistic Acceptability (correct style, grammar, ...)		Winograd, GLUE, Super GLUE, MMLU
Clustering & Topic Modeling*		Purity, Rand Index, Pointwise Mutual Information (PMI), Silhouette Index

One measurement, a few or all of them?

Despite their simplicity, language models are increasingly functioning as the foundation for almost all language technologies from question answering to summarization. But their immense capabilities and risks are not well understood. Holistic Evaluation of Language Models (HELM) is a living benchmark that aims to improve the transparency of language models.

1. **Broad coverage and recognition of incompleteness.** We define a taxonomy over the scenarios we would ideally like to evaluate, select scenarios and metrics to cover the space and make explicit what is missing.



2. **Multi-metric measurement.** Rather than focus on isolated metrics such as accuracy, we simultaneously measure multiple metrics (e.g., accuracy, robustness, calibration, efficiency) for each scenario, allowing analysis of tradeoffs.

	Metrics						
	Accuracy	Calibration	Robustness	Fairness	Bias	Toxicity	Efficiency
RAFT	✓	✓	✓	✓	✓	✓	✓
IMDB	✓	✓	✓	✓	✓	✓	✓
Natural Questions	✓	✓	✓	✓	✓	✓	✓
QuAC	✓	✓	✓	✓	✓	✓	✓
XSUM	✓				✓	✓	✓

3. **Standardization.** We evaluate all the models that we have access to on the same scenarios with the same adaptation strategy (e.g., prompting), allowing for controlled comparisons. Thanks to all the companies for providing API access to the limited-access and closed models and [Together](#) for providing the infrastructure to run the open models.

AI21labs

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Google

Meta

Microsoft



OpenAI



Yandex

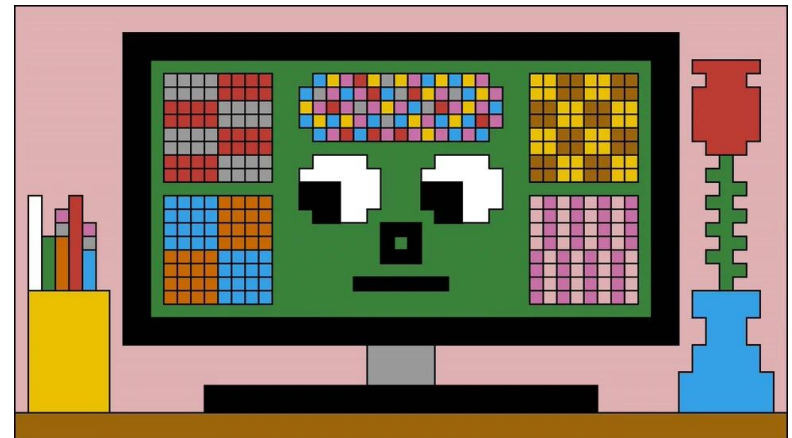
TOGETHER

<https://venturebeat.com/ai/stanford-debuts-first-ai-benchmark-to-help-understand-llms/>

<https://crfm.stanford.edu/helm/latest/>

Different Kind of Evaluation — Measuring Reasoning, Not Output Quality

- Every metric so far in this lecture (Precision/Recall, BLEU, ROUGE, perplexity, Kappa) scores how close an output is to a reference answer. They assume the task is well-defined and a "correct" output exists to compare against.
- ARC-AGI (François Chollet) asks a different question: not "is this output good?" but "did the system actually reason, or did it just memorize/pattern-match from training data?"
- Design idea: puzzle-style visual tasks that are easy for humans with only a few examples, but that can't be solved by recalling similar examples from a huge training set — there's no partial credit for having "seen something like this before."
- Why it matters: a model can score well on standard NLP benchmarks (high BLEU, high accuracy) through memorization or benchmark contamination, without demonstrating genuine generalization. High output-quality scores $\neq$ genuine reasoning ability.
- Takeaway: output-quality metrics (this lecture) and generalization/reasoning benchmarks (ARC-AGI) measure different things — a system can excel at one and fail badly at the other. Good evaluation means knowing which question you're actually asking.



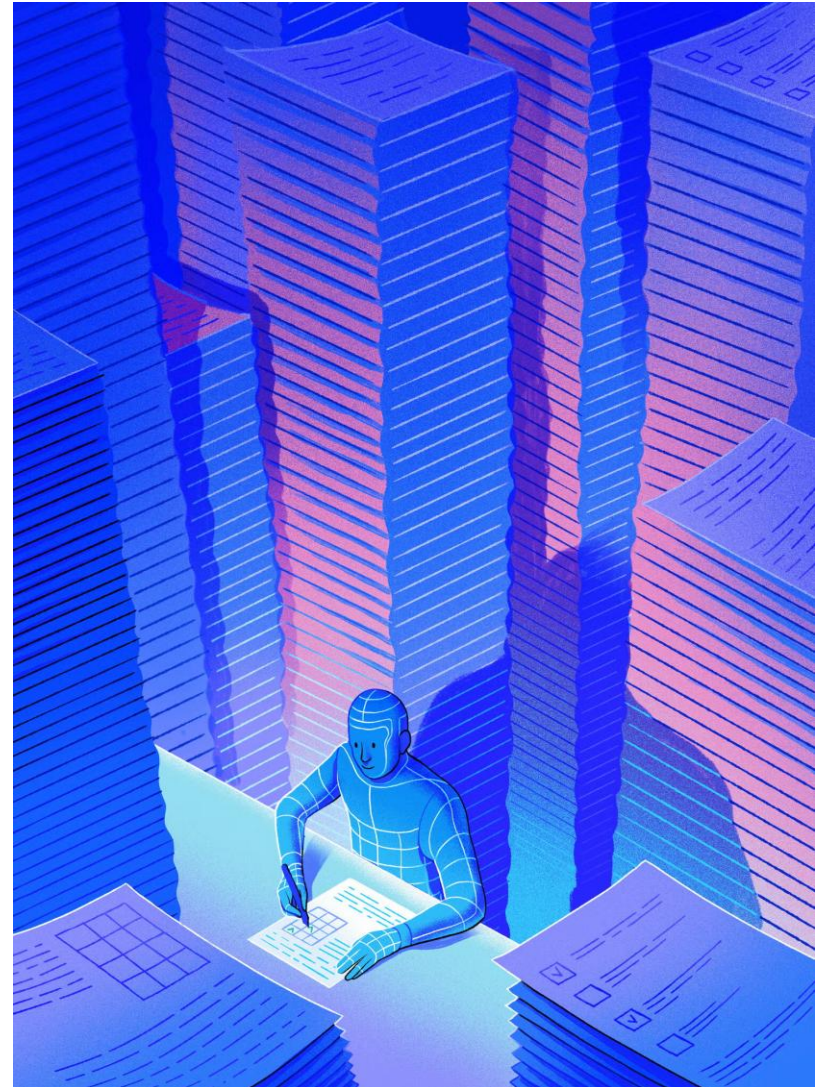
Source: Matteo Wong, "The Test That Stumps AI," The Atlantic (April 2025).

Humanity's Last Exam

Source: <https://www.nytimes.com/2025/01/23/technology/ai-test-humanitys-last-exam.html?smid=nytcore-ios-share&referringSource=articleShare>

- The creators of a new test called “Humanity’s Last Exam” argue we may soon lose the ability to create tests hard enough for A.I. models.
- Some of the smartest humans in the world are struggling to create tests that A.I. systems can’t pass.
- <https://lastexam.ai/> was recently created.

Example: Hummingbirds within Apodiformes uniquely have a bilaterally paired oval bone, a sesamoid embedded in the caudolateral portion of the expanded, cruciate aponeurosis of insertion of m. depressor caudae. How many paired tendons are supported by this sesamoid bone? Answer with a number.



Discussion Lecture: Quantitative Measurements



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Questions on the lecture?

Now let's form groups of 4



Discuss Exercises Lecture on Quantitative Measurements

Exercise 1

Suppose a medical search engine evaluates a corpus of $N = 100,000$ documents. Exactly 1,000 documents are relevant, while the remaining 99,000 irrelevant. A baseline classification model retrieves 500 documents in total, of which 400 are actually relevant.

- Construct the confusion matrix by calculating the exact counts for True Positives (tp), False Positives (fp), False Negatives (fn), and True Negatives (tn).
- Compute the model's **Accuracy**, **Precision**, and **Recall**.
- Compare these results to the "Snoogle.com" dummy baseline described in the lecture (a system that always retrieves 0 documents). Why does raw accuracy fail as an informative metric in unbalanced NLP and IR settings?
- In legal and medical domains, "lack of precision is cost, while lack of recall is risk". Using the generalized F_β formula from slide 16:

$$\text{with } F_\beta = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

- Should a legal compliance team select $\beta = 0.5$ or $\beta = 2.0$? Justify your choice based on how β weights precision versus recall.

Exercise 2

Consider the following reference and candidate sentences produced by a machine translation/summarization model:

- **Reference (R):** *"the doctor confirmed the patient was cured"*
- **Candidate 1 (C_1):** *"the patient was cured"*
- **Candidate 2 (C_2):** *"the doctor confirmed the patient was not cured"*
- Calculate unigram (p_1) and bigram (p_2) precision for both C_1 and C_2 using BLEU clipping rules against the reference.
- Explain how the **Brevity Penalty (BP)** in BLEU penalizes candidate C_1 . Why is a brevity penalty mathematically necessary when evaluating translation via precision-based n-gram overlap?
- Compute ROUGE-1 recall for C_2 against the reference. Notice that C_2 obtains a near-perfect ROUGE/BLEU score despite containing a critical factual contradiction ("not cured" vs. "cured"). Based on Tang et al. (2023), why can standard n-gram overlap metrics provide dangerous overconfidence in high-stakes domains (e.g., medicine or law)?
- How do semantic metrics like **COMET** or **BERTScore** resolve the structural blind spots of BLEU and ROUGE?

Exercise 3

Language models are evaluated across different levels of the data–information–knowledge continuum, from token probability modeling to pragmatic reasoning.

- What is **Perplexity** in terms of test set probability and entropy? If Model A achieves a perplexity of 16.5 on WikiText and Model B achieves 5.5, which model is a better predictor of the test distribution?
- Why do frontier chat models (such as GPT-4, Claude 3, and Gemini) report benchmark accuracies on MMLU, GSM8K, and HumanEval rather than publishing perplexity scores? Explain the impact of instruction tuning / RLHF on perplexity comparability.
- Consider the classic Winograd Schema pair:
 - *Sentence A: "The city councilmen refused the demonstrators a permit because they feared violence."*
 - *Sentence B: "The city councilmen refused the demonstrators a permit because they advocated violence."*
 - Why can standard syntactic parsing and lexical co-occurrence resolve neither pronoun reference without commonsense world knowledge? At which level of the continuum (Syntax, Semantics, Pragmatics, or Discourse) does this task reside?

Discussion Lecture: Quantitative Measurements



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Teaching note: Use today's metrics (precision/recall, perplexity, BLEU/ROUGE) as concrete ammunition — there's no single correct answer to these.

On Proxies: What Does a Number Know About Quality?

Every metric in this lecture replaces a question we care about (is this translation good? is this summary true?) with a question we can compute (how many n-grams overlap? how probable is this text?). The replacement is never exact.

Goodhart's law: when a measure becomes a target, it stops being a good measure. Models are now trained to maximise exactly these scores. What happens to a proxy once the thing it measures starts optimising for it?

Is “quality” a property of a text at all, or a relation between a text, a reader and a purpose? If the latter, can any reference-based metric measure it, even in principle?

On Perplexity and Truth: Can a Distribution Be Right?

Perplexity rewards a model for predicting what people write. If the corpus is full of confident falsehoods, a model with near-zero perplexity reproduces them perfectly. Is that an intelligent model or a perfect parrot?

Truth is not a property of a probability distribution. So what would a metric for “knowing” have to look like?

Is a benchmark exam (MMLU, the bar exam) such a metric, or just another corpus the model has learned to predict?

On the Gold Standard: When Humans Disagree, What Are We Measuring Against?

Every metric in this lecture compares a system to a human judgement: a reference translation, a relevance label, an annotated entity. Yet two annotators agree only about 80 to 90 percent of the time, and the same annotator disagrees with themselves on another day (slides 73 to 78).

If the gold standard is itself unreliable, what does it mean for a model to score 95 percent? Can a system be “better than human” on a task defined by human judgement?

Is disagreement between annotators noise to be averaged away, or information about the task that our metrics throw away?

Tutorial 03: Quantitative Measurements



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Questions?



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